

A T M Hasan

Senior Software Engineer | Java | Spring Boot | Microservices | AI/LLM (RAG)

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Professional Summary

Senior Software Engineer with 12+ years of experience in Java-based enterprise application development, backend systems, REST APIs, database-driven applications, and production troubleshooting. Experienced in leading development teams, API integration, SQL optimization, Agile delivery, and enterprise software support.

Professional Experience

Programming and ICT Instructor (Part-time), STEP IT Academy, Kuala Lumpur, Malaysia Jul 2025 – Present

- Deliver instructor-led Python training for beginner to intermediate learners, covering programming fundamentals, object-oriented programming, and introductory data analysis using pandas, NumPy, and Matplotlib.
- Guide students through hands-on coding exercises, assignments, and debugging to strengthen problem-solving and software development skills.
- Support certification preparation and structured learning pathways in software engineering, data analysis, and AI/ML.

Research Assistant (PhD Research – Privacy-Preserving RAG), UniKL–MIIT, Kuala Lumpur, Malaysia Jan 2025 – Present

- Conduct PhD research and hands-on implementation on privacy-preserving multimodal RAG for higher education, covering OCR, document parsing, retrieval, re-ranking, grounding, guardrails, and evaluation.
- Design and document experiments for LLM-powered document intelligence workflows involving text, scanned documents, and multimodal inputs.
- Contribute to technical reporting, results analysis, and manuscript preparation for research in RAG, multimodal AI, and privacy-aware knowledge systems.

Software Development Consultant (Part-time, Remote), Consumer Safety Watch, San Diego, CA, USA Mar 2018 – Dec 2024

- Design and develop backend services, REST APIs, and microservice-style components using Java and Spring Boot for enterprise applications.
- Improve system reliability and performance through profiling, SQL query optimization, production troubleshooting, log analysis, and monitoring.
- Collaborate with cross-functional teams in Agile environments, contributing to code reviews, technical documentation, and maintainable software delivery practices.

Software Development Manager (Remote), Next Bits, Buena Park, CA, USA Jul 2022 – Aug 2024

- Led a six-member engineering team delivering enterprise software solutions using Java, Spring Boot, Java Swing, and desktop application technologies.
- Managed database analysis, requirement planning, and modernization initiatives, including upgrading legacy systems to improve maintainability and long-term support.
- Oversaw Agile delivery, technical reviews, issue tracking, and engineering documentation to ensure software quality and timely product delivery.

Software Development Manager, CSL Technologies Limited, Dhaka, Bangladesh Dec 2014 – Mar 2021

- Supervised a five-member team building enterprise content management solutions using Java and OpenText Documentum.
- Led requirement analysis, project planning, Agile coordination, and product delivery for enterprise document management solutions.
- Automated and optimized document workflows to improve operational efficiency, compliance support, and turnaround time.

Key Projects

Business Management Solution (BZComposer)

Java | Spring Boot | MySQL | Team Leadership

- Designed and led the development of a business management solution to automate core business processes using Java, Spring Boot, and MySQL.
- Managed a team of 6 developers, delivered key features on time, and improved operational efficiency by 30%.

Enterprise Document Management System & Process Automation

Java | Spring Boot | OpenText Documentum | Oracle 11g

- Developed and automated core document-processing workflows for a large-scale banking operation using Java, Spring Boot, OpenText Documentum, and Oracle 11g.
- Improved document retrieval time by 40%, enhancing operational efficiency and document-management performance.

e-Invoicing Integration for MyInvois (Malaysia LHDNM)

PHP | MySQL | OAuth2 APIs | Digital Signatures | UBL-JSON

- Implemented end-to-end e-Invoicing integration for a retail POS system using PHP, MySQL, OAuth2 APIs, UBL-JSON, and digital-signature workflows.
- Enabled real-time invoice validation and compliance with LHDNM tax regulations.

Multi-Modal RAG for Higher Education

Python | OCR | FAISS/Chroma | RAG | Re-ranking

- Built a multimodal document question-answering system over PDFs, tables, and images using Python, OCR, vector search, and retrieval-augmented generation.
- Implemented ingestion, parsing, chunking, embedding indexing, retrieval, and re-ranking for grounded document-based answers.

Loan Prediction Model (Risk / Eligibility Scoring)

Python | scikit-learn | Cross-validation | Model Tuning

- Built a supervised machine learning model to predict loan approval and credit risk using historical applicant data.
- Created preprocessing pipelines for missing values, encoding, scaling, model comparison, and cross-validation using Python and scikit-learn.

Education

Doctor of Philosophy (Information Technology)

Jan 2025 – Present

Universiti Kuala Lumpur - Malaysian Institute of Information Technology (UniKL – MIIT)

Research Focus: Privacy-Preserving RAG for Higher Education: Synthetic Data, Minimization, and On-Prem Knowledge Stores.

Master of Science in Computer Science and Engineering

CGPA: 3.61/4.00

Jan 2019 – Dec 2020

United International University (UIU), Dhaka

Major in Software Engineering

Selected Publications

[1] M. Hasan and A. T. M. Hasan, “Using Titanic Dataset for Comprehensive Machine Learning Model Training,” International Journal of Innovative Science and Research Technology (IJISRT), Feb. 2025, doi: 10.5281/zenodo.14810217.

[2] A. T. M. Hasan, S. K. Ahammad Fahad, and S. M. Shariff, “Privacy-Preserving RAG for Higher Education: Synthetic Data, Minimization, and On-Prem Knowledge Stores,” manuscript under review, 2025. (Accepted for presentation at the 9th Global Higher Education Forum (GHEF 2025); full paper in publication process.)

Selected Certifications

- **IBM Cognitive Class:** Machine Learning with Python, Data Visualization with Python, Data Analysis with Python, Data Science Methodology, SQL and Relational Databases 101 (2025)
- **Building Scalable Java Microservices with Spring Boot and Spring Cloud**, Google Cloud (Oct 2024)
- **Python Programming Language**, Stanford Code in Place (Jun 2024)
- **Red Hat Certified System Administrator (RHCSA)**, Red Hat (Apr 2019)
- **Oracle Certified Professional, Java SE 6 Programmer (OCJP)**, Oracle (Mar 2015)

References

Dr. Shafiza Mohd Shariff Deputy Dean (Academic & Technology) Universiti Kuala Lumpur (UniKL–MIIT), KL, Malaysia Relationship: Academic	Dr. Syed Ali Chief Information Security Officer (CISO) Stealth ICT, LLC, Wyoming, United States Relationship: Professional
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